



Assessment For Learning

Subject: Mathematics

Grade Level: 10 D,E

Assessment Title: Sequences, Series and Trigonometry

Date: May 1/2026

Name of Student: _____

Outcomes Assessed:

1	Find missing terms by writing explicit formulas of arithmetic sequences and find sums of series. (Algebra II 13.2)
2	Find missing terms by writing explicit formulas of geometric sequences and find sums of series. (Algebra II 13.3)
3	Express the relationship between the unit circle and radian measure. (Algebra II 15.1)
4	Use the unit circle to define the trigonometric functions for all real numbers. (Algebra II 15.2)
5	Use a given trigonometric function to calculate the values of other trigonometric functions by means of the Pythagorean identity. (Algebra II 15.3)
6	Use the key features of the graphs to sketch transformed Sine, Cosine and Tangent functions (Notes and Worksheets posted on Google classroom.)
7	Use Sum and Difference trigonometric properties to calculate trigonometric values and simplify trigonometric expressions. (Notes and Worksheets posted on Google classroom.)
8	Use Double angle and Half angle trigonometric properties to calculate trigonometric values and simplify trigonometric expressions. (Notes and Worksheets posted on Google classroom.)

Feedback Summary:

Strength	
Improvement	
Student Response	

Marks Achieved

Total Marks

30

Percentage Grade

1. a_n is a geometric sequence where $a_2 = 0.2$ and $a_5 = 43.2$

a. Find a_1 .

[2]

b. Write an explicit formula of a_n .

[1]

c. Find:

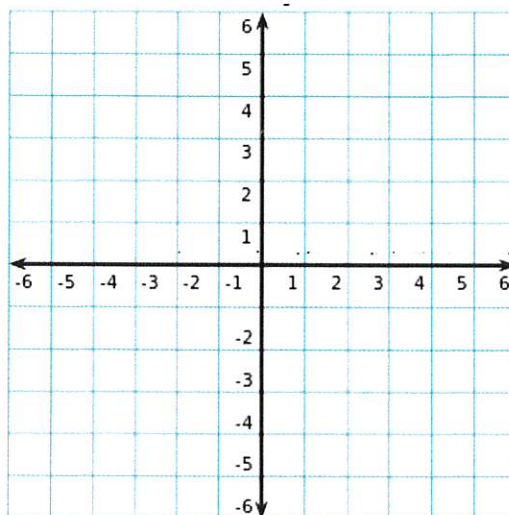
[2]

$$\sum_{n=1}^{n=15} a_n = ?$$

2. Given the coordinates of a point: $(3, -4)$

a. Plot the point on the coordinate system and construct the angle of rotation θ , associated with the given point.

[1]



b. Evaluate $\sin\theta$, $\cos\theta$ and $\tan\theta$.

[3]

3. Given $\cos\theta = -\frac{1}{2}$ where $\frac{\pi}{2} \leq \theta \leq \pi$

a. Calculate $\sin\theta$.

[2]

b. Calculate $\cos(2\theta)$ and $\tan(2\theta)$.

[3]

4. Given $\cos(2\theta) = \frac{\sqrt{3}}{2}$ where $0 \leq \theta \leq \frac{\pi}{2}$

[3]

Calculate $\sin(2\theta)$.

5. Without using a calculator, calculate the value of:

[3]

a. $\sin(-135) =$

b. $\tan\left(\frac{2\pi}{3}\right) =$

6. Simplify the below trigonometric expressions.

[3]

$$\cos(\pi - \theta) - \sin\left(\frac{\pi}{2} - \theta\right) =$$

7. Show that:

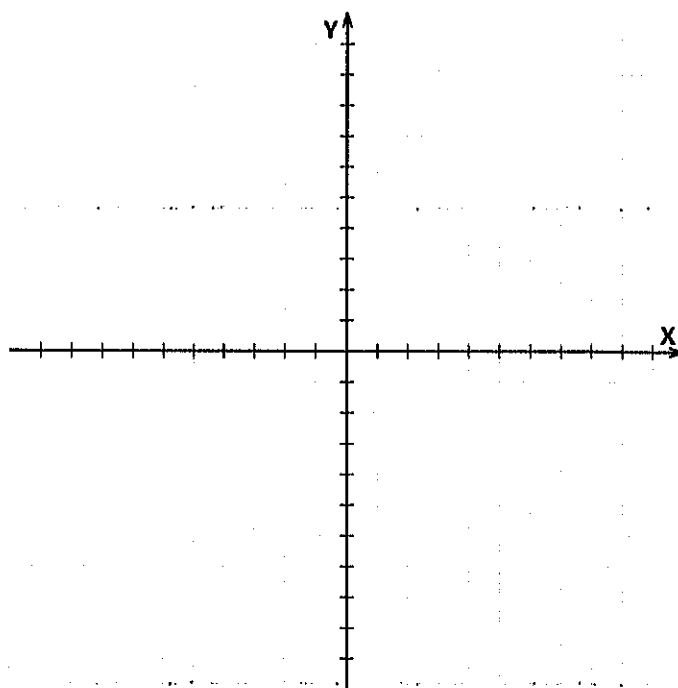
$$\frac{1}{\cos^2 x} - \tan^2 x = 1$$

[2]

8. Use the five-step method to sketch the function:

[5]

$$f(x) = \tan(x + \pi) - 1$$



End of AFL
Total 30 marks

Student Self-Assessment

Subject: Mathematics	Algebra 2_ AFL_ Term 3	Grade: 10 D, E	Dates: May 1, 2026
Module 13 and 14: Arithmetic sequences and series Module 15: Trigonometric Functions and Identities Additional Topics: Sum and Difference, Double Angle and Half Angle trigonometric properties.			
Student Name:	🍌 : I need help 🍌 : I believe I can do it 🍌 : I am confident I can do it and can show others		
	Before the lesson	After the lesson	I made progress
Learning Objectives	🍌 🍌 🍌 🍌 🍌 🍌	🍌 🍌 🍌 🍌 🍌 🍌	✓ ✗ ✓ ✗
1. Find missing terms by writing explicit formulas of arithmetic sequences and find sums of series.			13.2 Arithmetic Sequences and Series Pages 391 - 400
2. Find missing terms by writing explicit formulas of geometric sequences and find sums of series.			13.3 Geometric Sequences and Series Pages 401 - 410
3. Express the relationship between the unit circle and radian measure.			15.1 Angles of Rotation and Radian Measure Pages 437 - 444
4. Use the unit circle to define the trigonometric functions for all real numbers.			15.2 Define and Evaluate the Basic Trigonometric Functions Pages 445 - 452
5. Use a given trigonometric function to calculate the values of other trigonometric functions by means of the Pythagorean identity.			15.3 Use a Pythagorean Identity Pages 453 - 460
6. Use the key features of the graphs to sketch transformed Sine, Cosine and Tangent functions.			Notes and Worksheets posted on Google classroom.
7. Use Sum and Difference trigonometric properties to calculate trigonometric values and simplify trigonometric expressions.			Notes and Worksheets posted on Google classroom.